Network Mini-project

Ray Tsai

I. INTRODUCTION

Network growth models are central to our understanding of complex systems in the real world, ranging from the World Wide Web to biological and social networks. The Barabási–Albert (BA) model introduced preferential attachment as a mechanism for generating power-law degree distributions, while copying models propose an alternative microscopic rule in which new nodes replicate the connections of existing ones.

In this work we analyze the *all-or-none* model. At each growth step a new node is added and connected to a target chosen uniformly at random, and then it connects to *all* neighbors of the target with probability p, or to *none* of them with probability $1-p$. We show that despite its simplicity, the model produces power-law degree distributions with exponent $1/p$, an N-independent clustering coefficient $C = 3p/(1+2p)$, and logarithmic average distances, making it a small-world network for every $p \in (0,1)$.

The remainder of this paper is organized with Section II defining the model, Section III deriving the expected number of edges, Section IV deriving the degree distribution, Section V deriving the clustering coefficient, Section VI deriving the assortativity coefficient, and Section VII deriving the average shortest-path distance.

II. MODEL DEFINITION

The network begins with a single node ($N = 1$). At each growth step ($N \to N+1$):

1. A new node v is introduced.
2. A target node u is chosen uniformly at random from the existing N nodes.
3. An edge (v, u) is added.
4. With probability p, edges (v, w) are added for every neighbor w of u; with probability $1-p$, no further edges are added.

III. NUMBER OF EDGES

Let M denote the expected number of edges in the network, and write M_N for the expected number of edges when the network has N nodes. At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. The expected number of new edges is $1 + p\,k_u$, where k_u is the degree of u. Note that by the handshake lemma, the average value of k_u over all choices of u is $2M_N/N$. It follows that,

$$M_{N+1} = \left(1 + \frac{2p}{N}\right) M_N + 1. \tag{1}$$

This is a first-order linear recurrence. A particular solution of the form $\alpha_M N$ yields $\alpha_M = 1/(1-2p)$ for $p \neq 1/2$. The homogeneous solution is $\mathcal{A}_N = \Gamma(N+2p)/[\Gamma(1+2p)\,\Gamma(N)]$, satisfying $\mathcal{A}_{N+1}/\mathcal{A}_N = (N+2p)/N$. Imposing $M_1 = 0$ now yields

$$\boxed{M = \frac{1}{1-2p}\left(N - \frac{\Gamma(N+2p)}{\Gamma(1+2p)\,\Gamma(N)}\right)}, \tag{2}$$

for $p \neq 1/2$. Since $\Gamma(N+2p)/\Gamma(N) \sim N^{2p}$ as $N \to \infty$, two regimes emerge:

- *Sparse* ($p < 1/2$): $M \sim \frac{N}{1-2p}$ (linear).
- *Dense* ($p > 1/2$): $M \sim \frac{N^{2p}}{(2p-1)\,\Gamma(1+2p)}$ (superlinear).

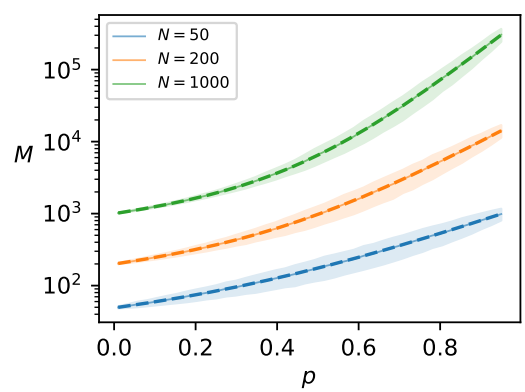

FIG. 1: Expected number of edges M (log scale) as a function of p for $N = 50, 200, 1000$ with 10^3 trials. Shaded regions show one standard deviation. The dashed lines are the analytical prediction (2), fitting the simulation results perfectly across all tested network sizes.

IV. DEGREE DISTRIBUTION

Denote by $f(k, i, N)$ the probability that node i has degree k when the network has N nodes. At step $N \to N+1$, degree k increases by one with probability $(1+pk)/N$, because either the node is chosen as target (probability $1/N$) or a neighbor of it is chosen and the

"all" event occurs (probability pk/N) The master equation for f is then

$$f(k,i,N+1) = \frac{1+p(k-1)}{N} f(k-1,i,N) + \left(1 - \frac{1+pk}{N}\right) f(k,i,N). \quad (3)$$

Assume that $p < 1/2$ so that the expected number of edges $\mathbb{E}[M]$ is linear in N, by the results of Section III. This allows us to seek the asymptotic degree distribution $P_k = \lim_{N\to\infty} n_k(N)/N$, where $n_k(N) = \sum_{i=1}^{N} f(k,i,N)$ is the expected number of nodes with degree k in the N-node network. Note that the new node enters with degree 1 with probability $1-p$ and with degree k with probability $p\, n_{k-1}(N)/N$ for $k \geq 2$. Hence, summing (3) over the N existing nodes and adding the new node yields

$$n_k(N+1) = \frac{1+p(k-1)}{N} n_{k-1}(N) + \left(1 - \frac{1+pk}{N}\right) n_k(N) + (1-p)\,\delta_{k,1} + \frac{p\, n_{k-1}(N)}{N}.$$

Substituting $n_k(N) = NP_k$, we have $(2+pk)\,P_k = (1+pk)\,P_{k-1} + (1-p)\,\delta_{k,1}$.. This yields $P_1 = (1-p)/(2+p)$ and, for $k \geq 2$, we have $P_k = \frac{1+pk}{2+pk} P_{k-1}$, which has the closed form

$$\boxed{P_k = \frac{1-p}{2+p} \prod_{j=2}^{k} \frac{1+pj}{2+pj}}, \quad k \geq 1.$$

In terms of the Gamma function,

$$P_k = \frac{(1-p)\,\Gamma(2+2/p)}{(2+p)\,\Gamma(2+1/p)} \cdot \frac{\Gamma(k+1+1/p)}{\Gamma(k+1+2/p)}. \quad (4)$$

As $k \to \infty$, we have $\Gamma(k+1+1/p)/\Gamma(k+1+2/p) \sim k^{-1/p}$, and so $P_k \sim \lambda_p k^{-1/p}$, where $\lambda_p = (1-p)\,\Gamma(2+2/p)/[(2+p)\,\Gamma(2+1/p)]$. The degree distribution thus follows a power law with exponent $1/p$, which matches the degree distribution of the BA model at $p = 1/3$.

V. TRIANGLES AND CLUSTERING

Let T denote the expected number of triangles in the network, and write T_N for the expected number of triangles when the network has N nodes. Triangles are created in an all-or-none network if and only if the "all" event occurs and $N \geq 2$. If the target u has degree k_u and there are e_u edges among its neighbors, the new node v forms k_u triangles of type (v,u,w_i) and e_u triangles of type (v,w_i,w_j), where w_i, w_j are distinct neighbors of u. Note that $\sum_u k_u = 2M_N$ and $\sum_u e_u = 3T_N$. Averaging over the target choice u thus yields

$$T_{N+1} = \left(1 + \frac{3p}{N}\right) T_N + \frac{2p\, M_N}{N}. \quad (5)$$

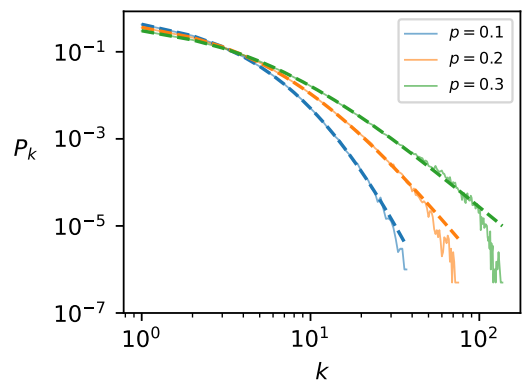

FIG. 2: Log-log scaled degree distribution P_k for $p = 0.1, 0.2, 0.3$ with $N = 10^3$, averaged over 10^3 realizations. Dashed lines show the analytical formula (4)

This linear recurrence has the same structure as (1). Substituting the exact edge formula and solving with $T_1 = 0$ gives, for $p \neq 1/2, 1/3$,

$$\boxed{T = \frac{2pN}{(1-2p)(1-3p)} + \frac{2\mathcal{A}_N}{1-2p} - \frac{2\mathcal{B}_N}{1-3p}}, \quad (6)$$

where $\mathcal{A}_N$ is as in (2) and $\mathcal{B}_N = \Gamma(N+3p)/[\Gamma(1+3p)\Gamma(N)]$. Thus two regimes emerge as $N \to \infty$:

- $p < 1/3$: $T \sim \frac{2p}{(1-2p)(1-3p)} N$ (linear).
- $p > 1/3$: $T \sim \frac{2}{\Gamma(1+3p)(3p-1)} N^{3p}$ (superlinear).

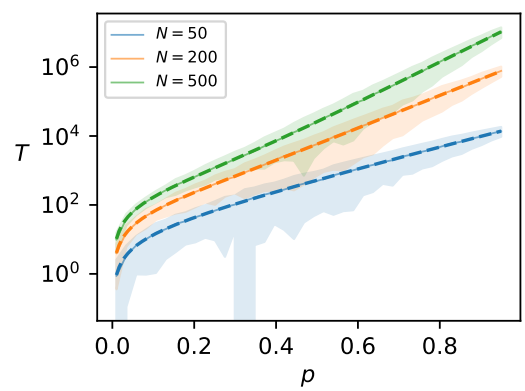

FIG. 3: Expected number of triangles T as a function of p for $N = 50, 200, 500$ with 10^3 trails. Shaded regions show one standard deviation. Dashed lines are the analytical prediction (6), fitting the simuation results perfectly across all tested networks sizes

We now derive the global clustering coefficient $C = 3T/W$, where $W = \frac{1}{2}\sum_i k_i(k_i-1)$ is the expected number of connected triples. Let $S_N = \sum_i k_i^2$, and note that

$W = (S_N - 2M_N)/2$. To derive a recurrence for S_N, note that when a node gains one link ($k \to k+1$), its contribution to S changes by $(k+1)^2 - k^2 = 2k+1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbor w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in S given target u is

$$\mathbb{E}[\Delta S \mid u] = (1-p)\cdot 1 + p(1+k_u)^2 + (2k_u+1) + p\sum_{w\sim u}(2k_w+1),$$

Notice that

$$\sum_u \sum_{w\sim u} k_w = \sum_i k_i^2 = S, \tag{7}$$

as each node w appears as a neighbor of exactly k_w other nodes. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, and (7) gives

$$S_{N+1} = \left(1 + \frac{3p}{N}\right) S_N + 2 + \frac{(4+6p)M_N}{N}. \tag{8}$$

Solving (8) by the same method used for (1) and (5) gives, for $p \neq 0, \frac{1}{2}, \frac{1}{3}$,

$$S = \frac{(6+2p)N}{(1-2p)(1-3p)} + \frac{(4+6p)\mathcal{A}_N}{p(1-2p)} - \frac{4(1+2p)\mathcal{B}_N}{p(1-3p)}, \tag{9}$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (6). Recall that $W = (S - 2M)/2$, so substituting (2) and (9) yields

$$W = \frac{2(1+2p)}{p}\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right],$$

Comparing with $3T$ from (6),

$$3T = 6\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right],$$

we see that W and $3T$ have the same bracketed term, and thus we get

$$\boxed{C = \frac{3p}{1+2p}}, \tag{10}$$

for all $p \in (0,1)$ as $N \to \infty$. In particular, this result is constant on N, which is large compared to the Erdős-Rényi random graph and thus a sign for the small-world property.

VI. ASSORTATIVITY

Let X, Y be the random variables representing the degrees of two nodes of an edge chosen uniformly at random from the network. The degree assortativity coefficient is

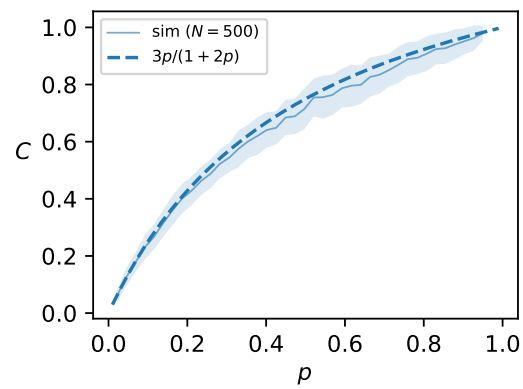

FIG. 4: Global clustering coefficient C as a function of p for $N = 500$ with 10^3 trials. Shaded region shows one standard deviation. The dashed line is the analytical prediction (10), fitting the simulation result tightly.

the Pearson correlation coefficient of the degree of pairs of adjacent nodes:

$$r = \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\mathbb{E}[X^2] - (\mathbb{E}[X])^2}.$$

Note that $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{|E|}\sum_{(u,v)\in E}\frac{1}{2}(k_u + k_v)$, $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \frac{1}{|E|}\sum_{(u,v)\in E}\frac{1}{2}(k_u^2 + k_v^2)$, and $\mathbb{E}[XY] = \frac{1}{|E|}\sum_{(u,v)\in E} k_u k_v$. Thus we have

$$r = \frac{\frac{1}{|E|}\sum_{(i,j)\in E} k_i k_j - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i + k_j)\right]^2}{\frac{1}{|E|}\sum_{(i,j)\in E}\frac{1}{2}(k_i^2 + k_j^2) - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i + k_j)\right]^2} = \frac{4MR - S^2}{2MQ - S^2}, \tag{11}$$

where $M = |E|$ and $S = \sum_i k_i^2$ are defined as before, $Q = \sum_i k_i^3$, and $R = \sum_{(i,j)\in E} k_i k_j$. Let Q_N and R_N denote the corresponding quantities when the network has N nodes.

We first derive a recurrence for Q_N. Note that when a node i gains one edge ($k_i \to k_i + 1$), its contribution to Q changes by $(k_i+1)^3 - k_i^3 = 3k_i^2 + 3k_i + 1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbor w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in Q given target u is

$$\mathbb{E}[\Delta Q \mid u] = (1-p)\cdot 1 + p(1+k_u)^3 + (3k_u^2 + 3k_u + 1) + p\sum_{w\sim u}(3k_w^2 + 3k_w + 1).$$

By the same argument as (7), $\sum_u \sum_{w\sim u} k_w^2 = \sum_w k_w^3 = Q$. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, $\sum_u k_u^3 = Q$, and (7)

yields

$$Q_{N+1} = \left(1 + \frac{4p}{N}\right) Q_N + 2 + \frac{(3+6p)S_N + (6+8p)M_N}{N}. \tag{12}$$

To derive a recurrence for R_N, note that when node i gains one edge, the product $k_i k_j$ on each incident edge (i,j) increases by k_j. If both endpoints gain degree simultaneously, the product increases by $k_i + k_j + 1$ instead. Write $\pi_i = \sum_{j\sim i} k_j$. We consider the two events separately.

"None" event. Only the target u gains degree 1, and the single new edge (v,u) contributes $k_v \cdot k_u^{\text{new}} = k_u + 1$. The total change is thus

$$\Delta R_{\text{none}} = \pi_u + k_u + 1.$$

"All" event. The target u and all its neighbors gain degree 1, so the set of nodes whose degree increases is $\{u\} \cup N(u)$. Each node i that gains degree 1 increases the product $k_i k_j$ by k_j on every incident edge (i,j), contributing π_i in total, summing over all such nodes yields

$$\pi_u + \sum_{w\sim u} \pi_w. \tag{13}$$

Also note that, for an edge (i,j) where both endpoints gain degree 1, the change in the product is $(k_i+1)(k_j+1) - k_i k_j = k_i + k_j + 1$, but (13) counts only $k_i + k_j$. Thus each such edge requires an additional $+1$ correction. The edges with both endpoints in $\{u\} \cup N(u)$ are the k_u edges (u,w) and the e_u edges among neighbors of u, so we need to add $k_u + e_u$ to the correction. For the new edges, the new node has degree $k_v = 1 + k_u$, so their total contribution to R is

$$(k_u+1)^2 + \sum_{w\sim u} (k_u+1)(k_w+1) = (k_u+1)(2k_u + 1 + \pi_u).$$

Combining all three parts, we get

$$\Delta R_{\text{all}} = (2+k_u)\pi_u + \sum_{w\sim u} \pi_w + 2k_u^2 + 4k_u + 1 + e_u.$$

Thus the expected change in R given target u is $\mathbb{E}[\Delta R \mid u] = (1-p)\,\Delta R_{\text{none}} + p\,\Delta R_{\text{all}}$. Note that

$$\sum_u \sum_{w\sim u} \pi_w = \sum_w k_w\, \pi_w = 2R, \tag{14}$$

as each edge $\{w,z\}$ contributes $k_w k_z$ once on each endpoint. Averaging over the uniform choice of u using (14), $\sum_u \pi_u = S$, $\sum_u k_u \pi_u = 2R$, $\sum_u e_u = 3T$, $\sum_u k_u = 2M$, and $\sum_u k_u^2 = S$ yields

$$R_{N+1} = \left(1 + \frac{4p}{N}\right) R_N + 1 + \frac{(1+3p)S_N + (2+6p)M_N + 3pT_N}{N}. \tag{15}$$

Both recurrences (12) and (15) share the homogeneous growth factor $(1 + 4p/N)$, with solution $\mathcal{C}_N = \Gamma(N+4p)/[\Gamma(1+4p)\,\Gamma(N)]$. Substituting the exact formulas for M_N, S_N, T_N from Sections III–V into the forcing terms and solving by the same method used for (1),(5), and (8) yields, for $p \neq 0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$,

$$Q_N = \frac{(26+22p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(12+36p+28p^2)\mathcal{A}_N}{2p^2(1-2p)} + \frac{4(3+12p+12p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_Q\, \mathcal{C}_N,$$

$$R_N = \frac{3(3+5p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(4+16p+18p^2)\mathcal{A}_N}{2p^2(1-2p)} + \frac{2(2+10p+15p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_R\, \mathcal{C}_N,$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (6), and λ_Q, λ_R are constants determined by the initial conditions $Q_1 = R_1 = 0$.

Since $\mathcal{C}_N \sim N^{4p}$, for $p < 1/4$, the homogeneous terms $\mathcal{A}_N$, $\mathcal{B}_N$, $\mathcal{C}_N$ are all sublinear and all five quantities M, S, T, Q, R grow linearly. Let

$$\alpha_Q = \frac{26+22p}{(1-2p)(1-3p)(1-4p)},\ \alpha_R = \frac{3(3+5p)}{(1-2p)(1-3p)(1-4p)}.$$

be the linear term coefficients of Q_N and R_N respectively. From (1) and (9) we can also see that $\alpha_M = 1/(1-2p)$ and $\alpha_S = (6+2p)/[(1-2p)(1-3p)]$. Substituting into (11) and simplifying, we get that, for $p < 1/4$, as $N \to \infty$,

$$\boxed{r \sim \frac{4\alpha_M \alpha_R}{2\alpha_M \alpha_Q - \alpha_S^2} = \frac{p(9-2p)}{(1+2p)(2-p)}}. \tag{16}$$

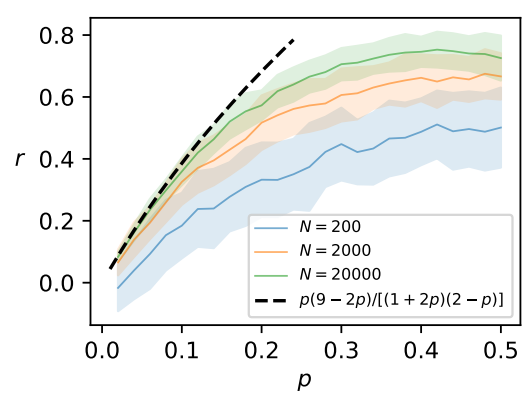

FIG. 5: Assortativity coefficient r as a function of p for $N = 200, 2000, 20000$ with 10^3 trials. Shaded regions show one standard deviation. The dashed line is the asymptotic prediction (16) for $p < 1/4$. The simuations slowly converges to our predicted curve, with higher values of p exhibiting even slower convergnece.

VII. AVERAGE SHORTEST-PATH DISTANCE

Let $D = \sum_{i<j} d(i,j)$ denote the expected sum of all pairwise shortest-path distances, and write D_N for this sum when the network has N nodes. The average shortest-path distance is then

$$\ell = \frac{2D_N}{N(N-1)}. \tag{17}$$

At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. Note that adding v does not shorten any existing pairwise distance, as any path through v can be rerouted through u or its neighbors without increasing length. Thus

$$D_{N+1} = D_N + \sum_j d(v,j).$$

Write $\sigma_u = \sum_{j\neq u} d(u,j)$. We now consider the two events separately.

"None" event. The new node v is connected only to the target u. Thus, $d(v,u) = 1$ and $d(v,j) = 1 + d(u,j)$ for every $j \neq u$, which yields

$$\sum_j d(v,j) = N + \sigma_u.$$

"All" event. The new node v is adjacent to the target u and all its neighbors. But then $N(u) = N(v)$, so any path from u to j can be rerouted through v by simply replacing the starting point u with v. Thus, $d(v,j) = d(u,j)$ for all $j \neq u$, and so

$$\sum_j d(v,j) = 1 + \sigma_u.$$

Averaging over the two events and the uniform choice of u, and noting that $\frac{1}{N}\sum_u \sigma_u = 2D_N/N$, we have

$$\sum_j d(v,j) = (1-p)N + p + \frac{2D_N}{N},$$

and so

$$D_{N+1} = \left(1 + \frac{2}{N}\right) D_N + (1-p)N + p. \tag{18}$$

The homogeneous solution is $\mathcal{D}_N = N(N+1)/2$, satisfying $\mathcal{D}_{N+1}/\mathcal{D}_N = (N+2)/N$. Set $D_N = \gamma_N \mathcal{D}_N$. Then (18) is reduced to

$$\gamma_{N+1} - \gamma_N = \frac{2[(1-p)N+p]}{(N+1)(N+2)} = \frac{2(2p-1)}{N+1} + \frac{2(2-3p)}{N+2}.$$

It follows from $\gamma_1 = 0$ that

$$\begin{aligned}\gamma_N &= \sum_{k=1}^{N-1} \frac{2(2p-1)}{k+1} + \frac{2(2-3p)}{k+2} \\ &= 2(2p-1)(H_N - 1) + 2(2-3p)\left(H_{N+1} - \frac{3}{2}\right),\end{aligned} \tag{19}$$

where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number. Since $H_N \sim \ln N$ as $N \to \infty$, we have $\gamma_N \sim 2(1-p)\ln N$, and so

$$D_N \sim (1-p)\, N(N+1) \ln N.$$

Substituting into (17) now yields

$$\boxed{\ell \sim 2(1-p)\ln N},$$

for all $p \in [0,1)$. At $p = 1$ the network is a complete graph with $\ell = 1$. The distance scales logarithmically for every with no phase transition, confirming the small-world property we established with the constant clustering coefficient from Section V.

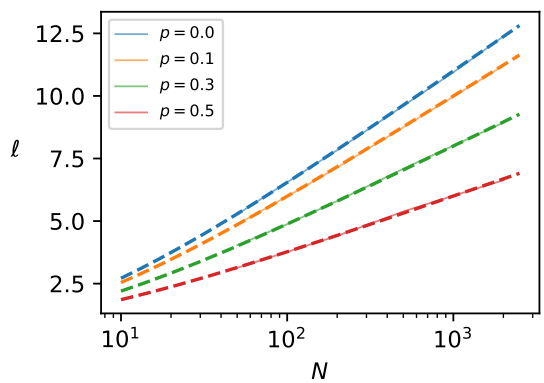

FIG. 6: Average shortest-path distance ℓ as a function of log scaled N for $p = 0.0, 0.1, 0.3, 0.5$. Dashed lines show the analytical prediction (19), fitting the simulation results perfectly across all tested p. Note that we use (19) in the figure instead of the asymptotic formula $2(1-p)\ln N$ to retain the constant correction terms.

VIII. CONCLUSION

We have studied the all-or-none network growth model and derived exact or asymptotic analytical expressions for five structural properties, all controlled by the single parameter p: (i) The expected number of edges transitions from linear in the sparse regime ($p < 1/2$) to superlinear in the dense regime ($p > 1/2$). (ii) The degree distribution follows a power law $P_k \sim k^{-1/p}$, reproducing the BA model exponent at $p = 1/3$. (iii) The global clustering coefficient converges to $C = 3p/(1+2p)$, independent of N and valid for all p. (iv) The assortativity coefficient converges to $r = p(9-2p)/[(1+2p)(2-p)]$ for $p < 1/4$. (v) The average shortest-path distance scales as $\ell \sim 2(1-p)\ln N$ for all $p < 1$, with no phase transition.

Together, results (iii) and (v) establish that the network exhibits the small-world property for all $p \in (0,1)$: logarithmic distances combined with large clustering.